

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Interactive DOM Manipulation using JavaScript

2. AIM

To demonstrate how to dynamically alter HTML content, CSS styles, and element attributes using JavaScript Document Object Model (DOM) methods.

3. OBJECTIVE

The objectives of this experiment are:

- To access DOM elements using `getElementById`.
- To attach event listeners to buttons to detect user clicks.
- To modify the inner text content of HTML elements.
- To change inline CSS properties dynamically.
- To set and modify HTML attributes dynamically.

4. THEORY

The Document Object Model (DOM) is a programming interface for web documents. It represents the page so that programs can change the document structure, style, and content. JavaScript interacts with the DOM to make web pages interactive by dynamically modifying HTML elements, manipulating their styles, and responding to user events like clicks.

5. ALGORITHM / PROCEDURE

1. Create a basic HTML structure containing a heading, a paragraph, and three buttons.
2. Apply CSS styling to the elements for a clean interface.
3. Select the heading and paragraph elements using JavaScript `getElementById()`.
4. Attach a click event listener to the 'Change Content' button to update the paragraph text.
5. Attach a click event listener to the 'Change Style' button to modify text color, background color, and font size.
6. Attach a click event listener to the 'Change Attribute' button to set a title attribute and modify the heading text.

6. CODE EXPLANATION

JavaScript is used to fetch specific DOM elements via `document.getElementById()`. Then, `addEventListener()` is chained onto button elements to listen for 'click' events. When triggered, the callback functions manipulate the DOM in various ways: `textContent` dynamically modifies the inner text, the `.style` property is used to override CSS styling such as color and `backgroundColor`, and `setAttribute()` programmatically adds an HTML attribute to an element.

7. PROGRAM

dom.html

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Example</title>

  <style>
    body {
      font-family: Arial;
      text-align: center;
      background-color: #f2f2f2;
      padding: 50px;
    }

    .box {
      background: white;
      width: 400px;
      margin: auto;
      padding: 30px;
      border-radius: 15px;
      box-shadow: 0 0 10px gray;
    }

    #message {
      padding: 15px;
      background-color: #e8f0ff;
      color: #333;
    }

    button {
      padding: 10px 20px;
      margin: 5px;
      border: none;
      border-radius: 5px;
      background-color: #007bff;
      color: white;
      cursor: pointer;
    }
  </style>
</head>

<body>

  <div class="box">

    <h1 id="heading">Student Information</h1>
```

```

<p id="message">
    Name: Vinay
</p>

<button id="contentBtn">
    Change Content
</button>

<button id="styleBtn">
    Change Style
</button>

<button id="attributeBtn">
    Change Attribute
</button>

</div>

<script>

    // Select HTML elements
    let heading = document.getElementById("heading");
    let message = document.getElementById("message");

    // Change content
    document.getElementById("contentBtn")
        .addEventListener("click", function() {

            message.textContent =
                "Name: Vinay | Branch: CSE (AI & ML)";

        });

    // Change style
    document.getElementById("styleBtn")
        .addEventListener("click", function() {

            heading.style.color = "blue";
            message.style.backgroundColor = "lightgreen";
            message.style.fontSize = "20px";

        });

    // Change attribute
    document.getElementById("attributeBtn")
        .addEventListener("click", function() {

            message.setAttribute(
                "title",
                "Student information updated"
            );

            heading.textContent =
                "Updated Student Information";
        });

```

```

    });

</script>

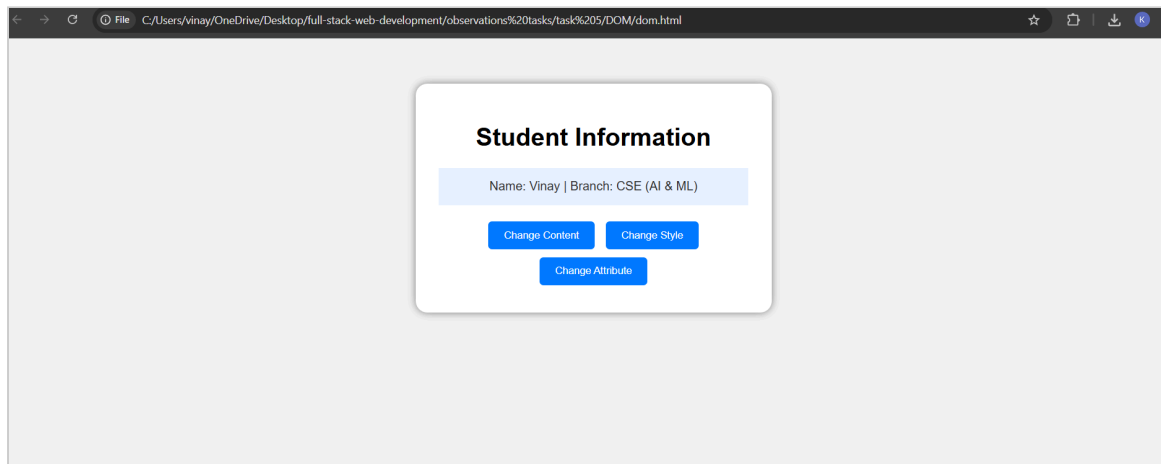
</body>
</html>

```

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Click 'Change Content'	Paragraph text updates	Paragraph text successfully updates to include Branch	Pass
TC-02	Click 'Change Style'	Element styles update	Heading turns blue and paragraph turns green	Pass
TC-03	Click 'Change Attribute'	Element attribute updates	Title attribute added and heading text changes	Pass

9. OUTPUT



10. RESULT

Thus, the dynamic manipulation of HTML content, CSS styles, and element attributes was successfully achieved using JavaScript DOM methods.